Problem: What is wrong with the following calculation?

$$\int_{-1}^{3} \frac{1}{x^2}\,dx = \frac{x^{-1}}{-1}\Big|_{-1}^{3} = -\frac{1}{3} - 1 = -\frac{4}{3}.$$

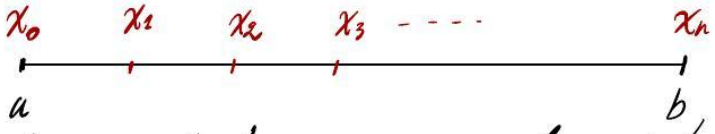
The Definite integral.

We know that using the limit

$$\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x.$$

we can compute the area under the curve $y = f(x)$.

Def. If f is a function defined for $a \le x \le b$, we divide the interval $[a,b]$ into n subintervals of equal width $\Delta x = \frac{b-a}{n}$



we denote $x_1^*, x_2^*, \dots, x_n^*$ by any sample points, x_i^*'s lies in the i-th subinterval.



lower Area. upper Area. Riemann Area.

Then, the definite integral of f from a to b

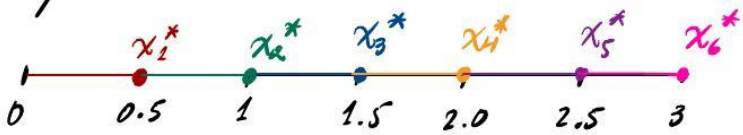
$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x.$$

provided that this limit and gives the same value for all possible choices of sample points.

Example. Evaluate the Riemann sum for $f(x) = x^3 - 6x$, $0 \le x \le 3$ with $n = 6$ subintervals and taking the sample points endpoints

to be right endpoints.

Eg.



$$\Delta x = \frac{3-0}{6} = 0.5.$$

then

$$R_6 = \sum_{i=1}^{6} f(x_i^*)\Delta x$$

$$= \Delta x \cdot \sum_{i=1}^{6} f(x_i^*)$$

$$= \frac{1}{2}\{f(0.5) + f(1) + f(1.5) + f(2) + f(2.5) + f(3)\}.$$

$$= \frac{1}{2}\{-2.875 - 5 - 5.625 - 4 + 0.625 + 9\} = -3.9375.$$

Thm 1. If f is a continuous function on $[a,b]$ or if f has only finite number of jump discontinuities then f is integrable on $[a,b]$; i.e, the definite integral $\int_a^b f(x)\,dx$ exists.

Thm 2. If f is an integrable function on $[a,b]$ then

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x.$$

where $\Delta x = \frac{b-a}{n}$; $x_i = a + i\Delta x$

Example. Express

$$\lim_{n\to\infty} \sum_{i=1}^{n} (x_i^3 + x_i \sin x_i)\Delta x$$

as an integral on the interval $[0,\pi]$.

Sln. Notice that

$$\int_0^{\pi} f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x = \lim_{n\to\infty} \sum_{i=1}^{n} (x_i^3 + x_i \sin x_i)\Delta x.$$

then $f(x_i) = x_i^3 + x_i \sin x_i$ i.e, $f(x) = x^3 + x\sin x$.

So, $$\lim_{n\to\infty} \sum_{i=1}^{n} (x_i^3 + x_i \sin x_i)\Delta x = \int_0^{\pi} (x^3 + x\sin x)\,dx.$$

Example. Evaluate $\int_0^3 (x^3-6x)\,dx$.

Sln. $f(x)= x^3-6x$; $\Delta x = \frac{b-a}{n} = \frac{3-0}{n} = \frac{3}{n}$.

$x_i = a + i\Delta x = 0 + \frac{3i}{n} = \frac{3i}{n}$ then.

$$\int_0^3 (x^3-6x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x.$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} f\left(\frac{3i}{n}\right)\frac{3}{n}$$

$$= \lim_{n\to\infty} \frac{3}{n} \sum_{i=1}^{n} \left(\left(\frac{3i}{n}\right)^3 - 6\left(\frac{3i}{n}\right)\right)$$

$$= \lim_{n\to\infty} \frac{3}{n} \sum_{i=1}^{n} \left(\frac{27\,i^3}{n^3} - \frac{18\,i}{n}\right)$$

$$= \lim_{n\to\infty} \left(\frac{3}{n} \sum_{i=1}^{n} \frac{27\,i^3}{n^3} - \frac{3}{n} \sum_{i=1}^{n} \frac{18\,i}{n}\right).$$

$$= \lim_{n\to\infty} \left(\frac{81}{n^4} \sum_{i=1}^{n} i^3 - \frac{54}{n^2} \sum_{i=1}^{n} i \right)$$

$$= \lim_{n\to\infty} \left(\frac{81}{n^4} \left(\frac{n(n+1)}{2}\right)^2 - \frac{54}{n^2}\cdot\frac{n(n+1)}{2}\right)$$

$$= \lim_{n\to\infty} \left(\frac{81}{n^4}\,\frac{n^2(n+1)^2}{4} - \frac{54}{n^2}\cdot\frac{n^2(1+1/n)}{2}\right)$$

$$= \lim_{n\to\infty} \left\{ \frac{81}{n^4}\cdot\frac{n^2\cdot n^2(1+1/n)^2}{4} - \frac{54}{2}\cdot(1+1/n)\right\}.$$

$$= \lim_{n\to\infty} \left\{ \frac{81}{4}\cdot\underbrace{(1+1/n)^2}_{1} - \frac{54}{2}\underbrace{(1+1/n)}_{1}\right\}. \qquad \frac{\#}{\infty}\longrightarrow 0$$

$$= \frac{81}{4} - \frac{54}{2} = \frac{81}{4} - 27 = -6.75.$$

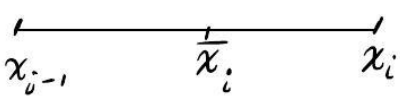
$\int_0^3 (x^3-6x)\,dx = F(3) - F(0)$ if F is an antiderivative.

$F(x) = \frac{x^4}{4} - 3x^2$. Thus.

$$\int_0^3 (x^3-6x)\,dx = F(3) - F(0) = \frac{3^4}{4} - 0 = \frac{81}{4}.$$

The midpoint rule:

$$\int_a^b f(x)\,dx \approx \sum_{i=1}^{n} f(\bar{x}_i)\,\Delta x \qquad \text{where } \Delta x = \frac{b-a}{n}$$



$$\bar{x}_i = \frac{x_{i-1} + x_i}{2}.$$

Properties of the Definite integral.

i) $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx.$

ii) $\int_a^a f(x)\,dx = 0$

iii) $\int_a^b c \cdot f(x)\,dx = c \cdot \int_a^b f(x)\,dx.$

iv) $\int_a^b (f(x) \pm g(x))\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx.$

Example. Compute $\int_0^1 (4 + 3x^2)\,dx.$

Sln.

$$\int_0^1 (4+3x^2)\,dx = \int_0^1 4\,dx + \int_0^1 3x^2\,dx$$

$$= 4\int_0^1 1\,dx + 3\int_0^1 x^2\,dx$$

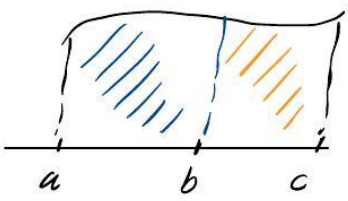
$$= 4x\Big|_0^1 + 3 \cdot \frac{x^3}{3}\Big|_0^1$$

$$= 4x\Big|_0^1 + x^3\Big|_0^1$$

$$= (4-0) + (1-0) = 5.$$

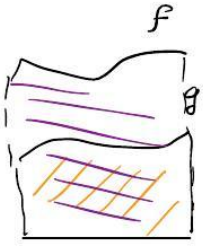
More Properties.

v) $\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx$



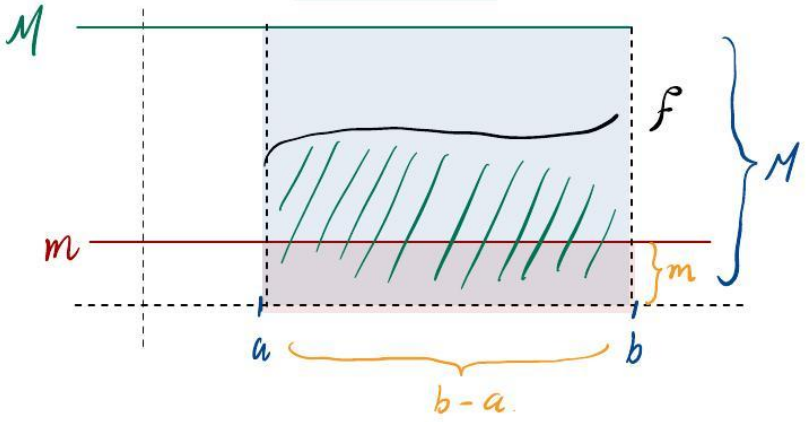
vi) If $f(x) \geq 0 \quad a \leq x \leq b$ then $\int_a^b f(x)\,dx \geq 0.$

vii) If $f(x) \geq g(x)$ for $a \leq x \leq b$ $\int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx.$



viii). If $m \le f(x) \le M$ for $a \le x \le b$ then

$$m(b-a) \le \int_a^b f(x)\,dx \le M\cdot(b-a).$$



How we can compute antiderivatives?

Goal: Evaluate $\int_a^b f(x)\,dx$ using FTC P2.

Indefinite Integrals.

We need a convenient notation for antiderivatives that makes them easy to work with.

$$\int f(x)\,dx = F(x) \quad \text{means} \quad F'(x) = f(x).$$

The notation $\int f(x)\,dx$ is traditionally used for an antiderivative of f and is called an indefinite integral.

$\int_a^b f(x)\,dx$ ⟶ Area under the curve $y = f(x)$.

but.

$\int f(x)\,dx$ ⟶ Antiderivative of $f(x)$.

For example: $\int x^2\,dx = \frac{x^3}{3} + C$.

b/c $\frac{d}{dx}\left(\frac{x^3}{3} + C\right) = x^2$.

Rmk: You should distinguish carefully between definite and indefinite integrals. A definite integral $\int_a^b f(x)\,dx$ is a <u>number</u> and an

indefinite integral $\int f(x)\,dx$ is a function (or family of functions).

The connection between them is given by.

$$\int_a^b f(x)\,dx = \underbrace{\int f(x)\,dx}_{\text{an antiderivative}} \Big|_a^b$$

Example. $\int \sec^2 x\,dx = \tan x + C$ b/c $\frac{d}{dx}(\tan x + C) = \sec^2 x$.

Table of indefinite integrals.

1). $\int c f(x)\,dx = c\int f(x)\,dx$.

2. $\int k\,dx = kx + C$.

3). $\int (f(x) \pm g(x))\,dx = \int f(x)\,dx \pm \int g(x)\,dx$.

4). $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$.

5). $\int \sin x\,dx = -\cos x + C$.

6). $\int \sec^2 x\,dx = \tan x + C$.

7). $\int \cos x\,dx = \sin x + C$.

8). $\int \csc^2 x\,dx = -\cot x + C$.

9). $\int \sec x \tan x\,dx = \sec x + C$.

10). $\int \csc x \cot x\,dx = -\csc x + C$.

Example. $\int \frac{1}{x^2}\,dx = \int x^{-2}\,dx = \frac{x^{-2+1}}{-2+1} + C = -\frac{1}{x} + C$.

Example. Find

i). $\int (10x^4 - 2\sec^2 x)\,dx$.

ii). $\int \frac{\cos\theta}{\sin^2\theta}\,d\theta$.

Sln. i) $\int (10x^4 - 2\sec^2 x)\,dx = 10\underbrace{\int x^4\,dx}_{\text{using 4.}} - 2\underbrace{\int \sec^2 x\,dx}_{\text{using 6.}}$.

$$= 10\,\frac{x^{4+1}}{4+1} - 2\cdot\tan x + C.$$

$$= 2x^5 - 2\tan x + C.$$

$$\frac{d}{dx}\left(2x^5 - 2\tan x + C\right) = 10x^4 - 2\sec^2 x.$$

ii). $\int \frac{\cos\theta}{\sin^2\theta}\,d\theta = \int \frac{\cos\theta}{\sin\theta}\cdot\frac{1}{\sin\theta}\,d\theta = \int \underbrace{\cot\theta\cdot\csc\theta\,d\theta}_{\text{using 10.}} = -\csc\theta + C.$

Example. Find $\int_0^{12}(x - 12\sin x)\,dx$.

Sln. $\int (x - 12\sin x)\,dx = \int x\,dx - 12\int \sin x\,dx.$

$$= \frac{x^2}{2} - 12(-\cos x) + C.$$

$$= \frac{x^2}{2} + 12\cos x + C.$$

$$\int_0^{12}(x - 12\sin x)\,dx = \left(\frac{x^2}{2} + 12\cos x + C\right)\Big|_0^{12}$$

$$= \frac{12^2}{2} + 12\cos 12 + C - \left(\frac{0^2}{2} + 12\cos 0 + C\right)$$

$$= 72 + 12\cos 12 + \cancel{C} - 12 - \cancel{C}$$

$$= 60 + 12\cos 12.$$

Problem: Evaluate $\int_1^9 \frac{2t^2 + t^2\sqrt{t} - 1}{t^2}\,dt.$